# ARIMA Time Series Forecasting: Pakistan Rainfall Analysis & Flood Prediction

COMP 4581 - Algorithms for Data Science

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## Abstract

In this project, I applied ARIMA and SARIMA time series forecasting models to 116 years of monthly rainfall data from Pakistan (1901-2016). The goal was to analyze monsoon patterns and see whether algorithmic forecasting could work as an early warning system for abnormal precipitation events. Five models were compared - ARIMA, SARIMA, SARIMAX (with ENSO ocean temperature data), a historical average baseline, and a last-year-repeat baseline - using a train/test split of 1901-2010 and 2011-2016. SARIMA achieved the best performance with an RMSE of 14.06 mm by capturing the 12-month monsoon seasonality that basic ARIMA misses. Adding Pacific Ocean surface temperature anomalies (ENSO) as an exogenous variable yielded only a 0.1% improvement, suggesting that SARIMA already captures the seasonal signal effectively at the monthly level. I was surprised that the algorithm couldn't accurately predict the exact monsoon rainfall in 2022, but I found that the confidence intervals still provide a useful baseline for anomaly detection - which could support real-time flood monitoring systems.


## 1. Introduction

Pakistan is my home country, and climate change is something I care about on a personal level. I wanted to do meaningful research work for my country with this project. The geography - glaciers up north, flat plains in Sindh and Punjab - makes Pakistan especially vulnerable to monsoon flooding. The 2022 monsoon season was catastrophic: rainfall reached approximately ten times normal levels, displacing over 30 million people and causing an estimated $30 billion in damages [1].

And the problem keeps getting worse. Pakistan's population has grown from roughly 45 million in 1960 to over 250 million in 2024, with urbanization doubling to nearly 40% [2]. Meanwhile, forest cover has declined from 6.5% to 4.7% of total land area [2], reducing the natural capacity to absorb excess rainfall. These trends mean that even moderate increases in monsoon intensity translate into disproportionately larger human and economic costs.

The main question I wanted to answer is whether ARIMA and its seasonal variants can actually learn from over a century of rainfall history and give us useful predictions about future monsoon patterns. While perfect prediction of extreme events may be impossible, establishing a quantitative baseline of "normal" rainfall allows anomaly detection: when actual measurements significantly exceed the model's confidence interval, that deviation itself becomes the early warning signal.

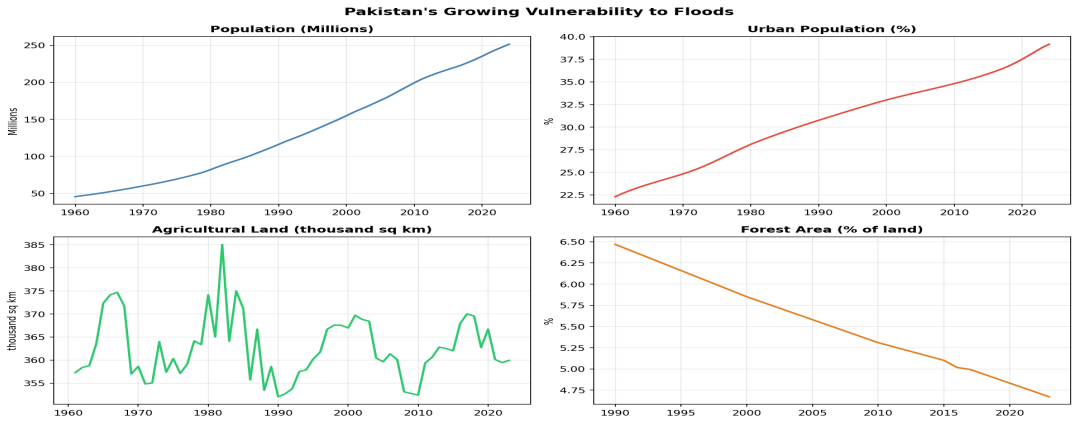


*Figure 1: Pakistan's growing vulnerability - population, urbanization, agricultural land, and forest cover trends (World Bank, 1960-2024).*

# 2. Algorithm Background

## 2.1 ARIMA

ARIMA stands for Auto-Regressive Integrated Moving Average. It has three main components. The Auto-Regressive (AR) component of order p uses p past observations to predict the current value - essentially capturing momentum in the data. The Integrated (I) component of order d applies differencing to achieve stationarity, removing trends that would violate the model's assumptions. The Moving Average (MA) component of order q uses q past forecast errors to correct predictions, allowing the model to learn from its own mistakes.

An ARIMA(p, d, q) model is fitted by estimating parameters that maximize the likelihood of the observed data, typically using the Kalman filter. I used AIC (Akaike Information Criterion) to pick between different model configurations. It basically balances how well the model fits against how complex it is, to avoid overfitting.

## 2.2 SARIMA

Seasonal ARIMA extends the base model with additional parameters (P, D, Q) that operate at a seasonal lag s. For monthly data with annual seasonality, s = 12. This means the model considers not just what happened 1-2 months ago, but also what happened 12, 24, and 36 months ago - capturing recurring annual patterns like Pakistan's monsoon cycle. A SARIMA model is written as SARIMA(p, d, q)(P, D, Q, s).

For Pakistan rainfall, this distinction is critical. Monthly rainfall in July is far more correlated with previous Julys than with the preceding June. Basic ARIMA, lacking seasonal awareness, cannot capture this relationship and consequently produces significantly worse forecasts.

### 2.3 SARIMAX

SARIMAX extends SARIMA by incorporating exogenous (external) variables alongside the time series. In this project, the Oceanic Nino Index (ONI) - a measure of sea surface temperature anomalies in the tropical Pacific - was used as the exogenous variable. The El Nino-Southern Oscillation (ENSO) cycle is known to influence South Asian monsoon intensity, with La Nina events (negative ONI) tending to increase rainfall in Pakistan [3].

### 2.4 Complexity

The Kalman filter used for ARIMA parameter estimation runs in O(n) per iteration, where n is the number of observations. Total fitting time depends on the number of iterations required for convergence and the number of parameters. SARIMAX adds minimal overhead beyond SARIMA - one additional coefficient for the exogenous variable. Empirical measurements confirmed roughly linear scaling from 120 to 1,320 observations.

## 3. Data

Finding the right data took some effort. I went through a few sources before settling on three datasets:

**Kaggle Pakistan Rainfall (primary):** Monthly average rainfall in millimeters for Pakistan, 1901-2016. 1,392 observations across 116 years with no missing values. Source: Kaggle, originally derived from meteorological records [4].

**World Bank Climate Indicators (contextual):** Population, urbanization, agricultural land, and forest cover for Pakistan, 1960-2024. Used to establish the socioeconomic context for why flood forecasting matters [2].

**NOAA Oceanic Nino Index (external variable):** Monthly three-month running mean sea surface temperature anomalies in the Nino 3.4 region, 1950-2017. Used as the exogenous variable in SARIMAX to test whether ocean temperature data improves forecasts [5].

It was hard finding data on Kaggle that was both reliable and had enough coverage, but this one turned out to be very clean - no missing values across all 116 years. The rainfall dataset exhibited a clear seasonal pattern with July and August averaging 56.6 mm and 51.8 mm respectively - approximately 2-3 times any other month. The Augmented Dickey-Fuller test confirmed stationarity (test statistic = -6.70, $p < 0.001$), establishing $d = 0$ for ARIMA. The ACF plot showed a strong spike at lag 12, confirming yearly seasonality and motivating the use of SARIMA over basic ARIMA.

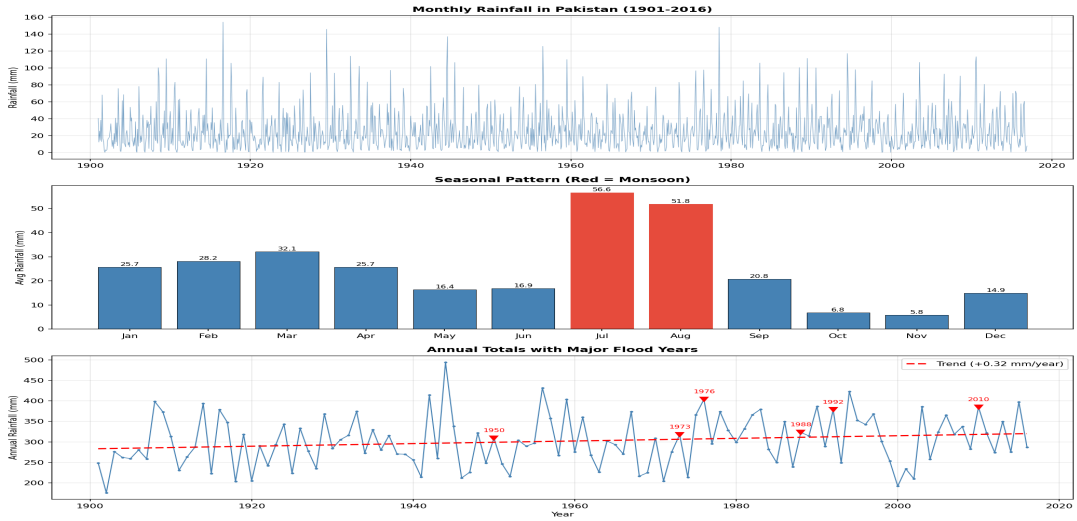


*Figure 2: Monthly rainfall time series (top), seasonal pattern with monsoon highlighted in red (middle), and annual totals with major flood years marked (bottom).*

# 4. Methodology

## 4.1 Train/Test Split

The dataset was split temporally: 1901-2010 (1,320 months) for training and 2011-2016 (72 months) for testing. This ensures the model is evaluated on data it has never seen, simulating a real forecasting scenario.

## 4.2 Stationarity and Parameter Selection

The ADF test confirmed stationarity (d = 0). ACF and PACF plots were used to guide initial parameter selection, with the strong lag-12 spike in the ACF confirming the need for seasonal modeling. A grid search over five ARIMA configurations and five SARIMA configurations was performed, selecting the model with the lowest AIC.

## 4.3 Model Configurations Tested

Basic ARIMA: (1,0,1), (2,0,1), (1,0,2), (2,0,2), (3,0,1). Best: ARIMA(2,0,2) with AIC = 11,578.

SARIMA: five seasonal configurations with period s = 12. Best: SARIMA(1,0,2)(1,1,1,12) with AIC = 10,967 - a drop of approximately 600 from the best basic ARIMA, indicating substantial improvement from the seasonal component.

SARIMAX: the best SARIMA configuration was augmented with the ONI index as an exogenous variable, using the overlapping period 1950-2010 for training and 2011-2016 for testing.

## 4.4 Baselines

Two naive baselines were included to verify that the algorithm adds genuine predictive value. The Historical Average baseline predicts each month using the average rainfall for that month across all

training years. The Last Year Repeat baseline assumes next year will be identical to the most recent training year (2010).

### 4.5 Evaluation Metrics

Models were compared using Root Mean Squared Error (RMSE), which penalizes large errors more heavily, and Mean Absolute Error (MAE), which gives a more interpretable average deviation. Lower values indicate better performance for both metrics.

## 5. Results

### 5.1 Model Comparison

| Model | RMSE (mm) | MAE (mm) | Notes |
|---|---|---|---|
| ARIMA(2,0,2) | 16.64 | 13.48 | No seasonality |
| **SARIMA(1,0,2)(1,1,1,12)** | **14.06** | **10.98** | **Best overall** |
| SARIMAX (+ ENSO) | 13.70 | 10.62 | +0.1% vs SARIMA |
| Historical Average | 14.09 | 11.01 | Naive baseline |
| Last Year Repeat | 27.63 | 18.38 | Worst performer |

*Table 1: Model performance on the test set (2011-2016). SARIMA achieves the lowest RMSE.*

SARIMA outperformed all other models, achieving an RMSE of 14.06 mm compared to 16.64 mm for basic ARIMA - a 15% reduction attributable entirely to the seasonal component. The Last Year Repeat baseline performed worst at 27.63 mm, confirming that the algorithm provides genuine predictive value beyond simple memorization.

The Historical Average baseline (14.09 mm) performed surprisingly close to SARIMA, which is expected: for a highly regular seasonal pattern, the long-run average is a strong predictor. However, SARIMA has the advantage of adapting to recent trends, while the historical average treats all years equally.

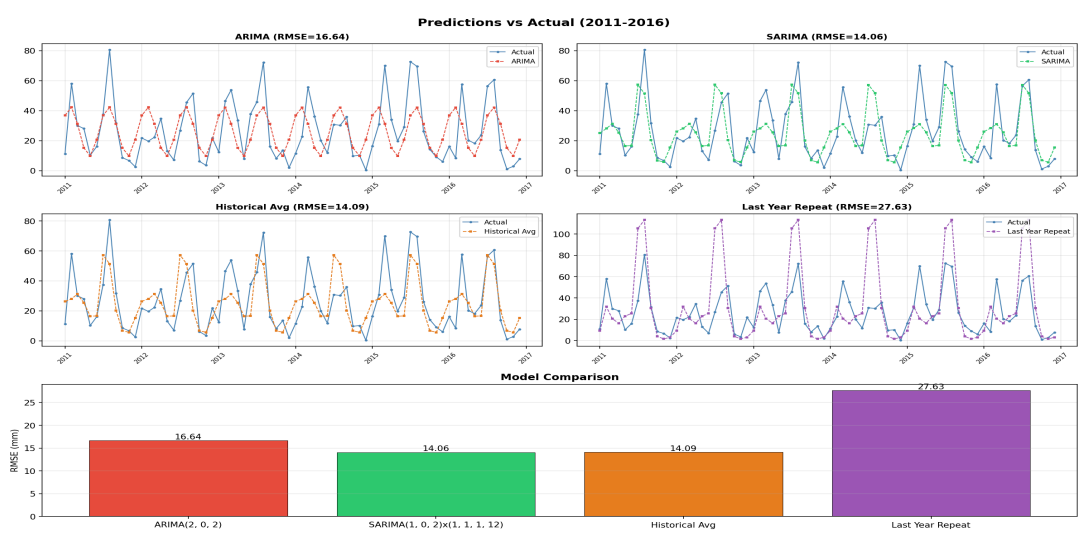

*Figure 3: Model predictions vs actual rainfall (2011-2016) for all four models, with RMSE comparison.*

## 5.2 ENSO Analysis

The correlation between the ONI and monsoon-season (July-August) rainfall was r = -0.258, confirming the expected negative relationship: La Nina events are associated with heavier monsoon rainfall in Pakistan. However, SARIMAX incorporating ONI as an exogenous variable improved RMSE by only 0.1% compared to SARIMA alone.

This minimal improvement suggests that SARIMA already captures the seasonal rainfall pattern effectively from historical data. ENSO likely influences monsoon intensity at a broader seasonal scale - determining whether an entire monsoon season will be above or below average - rather than affecting individual monthly values. This finding indicates that ENSO data may be more useful in seasonal forecasting models rather than monthly ARIMA variants.

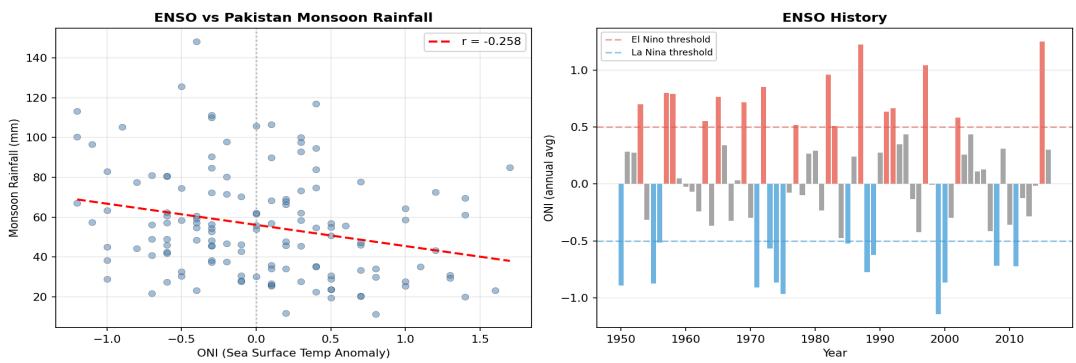


*Figure 4: ENSO vs Pakistan monsoon rainfall scatter plot (left) and ONI history showing El Nino/La Nina cycles (right).*

## 5.3 Forecasting and the 2022 Context

The SARIMA model, retrained on all available data (1901-2016), forecast approximately 57 mm for July 2022 with a 95% confidence interval of 24-89 mm. The actual 2022 monsoon season brought roughly ten times normal rainfall - far exceeding the model's upper confidence bound.

This result illustrates both the limitation and the utility of ARIMA-based forecasting. The model cannot predict unprecedented events that fall outside the historical distribution. However, the confidence interval provides a quantitative definition of "normal." In a real-time monitoring context, actual readings that massively exceed the upper bound serve as an objective anomaly signal - precisely the kind of trigger needed for early warning systems.

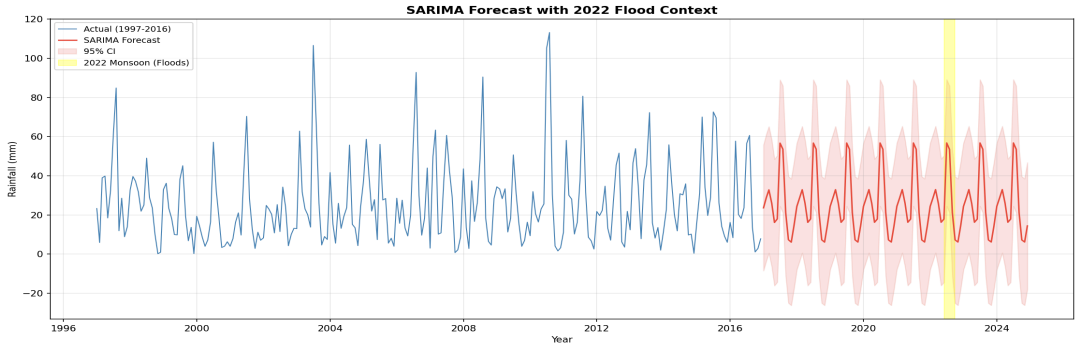


*Figure 5: SARIMA forecast (2017-2024) with 95% confidence interval. The yellow band marks the 2022 monsoon season.*

## 5.4 Monsoon Variability

Analysis of July-August rainfall totals across 116 years revealed that monsoon variability - measured by the 20-year rolling standard deviation - decreased during the mid-20th century and has been increasing in recent decades. This trend toward less predictable monsoon seasons is consistent with broader climate change research and suggests that the historical baseline ARIMA relies on may become less reliable over time as climate patterns shift.

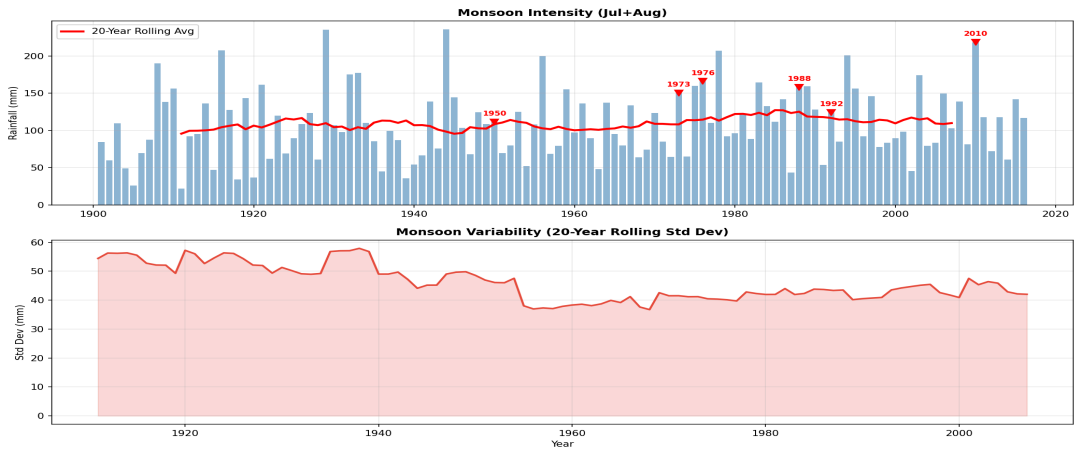


*Figure 6: Monsoon intensity (top) and variability (bottom) over 116 years, with major flood years marked.*

## 5.5 Complexity Analysis

Empirical measurements of SARIMA training time across data sizes from 120 to 1,320 observations confirmed roughly linear scaling, consistent with the theoretical O(n) per-iteration cost of the Kalman filter. Training on the full 110-year dataset required approximately 7 seconds. SARIMAX added negligible overhead beyond SARIMA.

# 6. Discussion

The biggest takeaway from this project is that seasonal modeling is essential for Pakistan rainfall. The 600-point AIC reduction from ARIMA to SARIMA - and the corresponding 15% RMSE improvement -

confirm that the monsoon cycle is by far the strongest signal in the data. So any forecasting system for Pakistan would need to account for seasonality as the number one priority.

I was honestly surprised how close SARIMA and the Historical Average performed. It made me question whether the algorithm was actually doing anything beyond just averaging. The answer is yes, but the reasons are kind of subtle. SARIMA adapts to recent trends and can pick up on gradual shifts in monsoon patterns that a static average would miss. As climate change keeps altering rainfall distributions, this adaptability matters more and more.

The ENSO result - minimal improvement at the monthly level - is itself a meaningful finding. It suggests that the relationship between ocean temperatures and Pakistan rainfall operates at a seasonal or inter-annual scale rather than a monthly one. Future work incorporating ENSO at the seasonal forecasting level, rather than as a monthly exogenous variable, may yield better results.

I think the most important thing I learned is that ARIMA's real value isn't about predicting exact rainfall numbers - it's about detecting when something abnormal is happening. It's still pretty cool that the confidence interval and standard deviation can be used as a trigger to kick in an emergency response before things get worse. In a country where 30 million people were displaced by floods in one summer, even a small improvement in early detection could save a lot of lives.

## 7. Limitations

At the end of the day, ARIMA is a linear model - it can only learn from what it's seen before. It can't model the complex climate physics that drive extreme events. The 2022 floods, driven by unprecedented atmospheric conditions potentially linked to accelerating climate change, fall outside the range of anything in the 116-year training set.

Only one external variable (ENSO) was tested. Other potentially relevant factors include the Indian Ocean Dipole, glacial melt rates, soil moisture conditions, and land use changes. The dataset also provides only a single national-level rainfall measurement; regional analysis (separating Sindh, Punjab, Balochistan, and KPK) could reveal important spatial patterns.

Finally, the increasing monsoon variability observed in recent decades suggests that the stationarity assumption underlying ARIMA may weaken over time as climate change accelerates.

## 8. Future Work

There are a few directions that could build on what I did here. Ensemble models combining ARIMA with LSTM neural networks could leverage both the interpretability of ARIMA and the nonlinear modeling capacity of deep learning. Seasonal-scale forecasting - predicting total monsoon-season rainfall rather than individual monthly values - may better capture the ENSO relationship. Incorporating satellite-derived precipitation estimates and soil moisture data could enable real-time monitoring. Finally, regional analysis at the provincial level would provide more actionable forecasts for local disaster management authorities.

## 9. Conclusion

In this project I applied ARIMA, SARIMA, and SARIMAX to 116 years of Pakistan rainfall data, comparing five models across three datasets. SARIMA(1,0,2)(1,1,1,12) achieved the best performance by capturing the monsoon's 12-month seasonal cycle. ENSO ocean temperatures confirmed the expected La Nina-rainfall relationship but provided minimal forecasting improvement at the monthly level. The model's confidence intervals, while unable to predict unprecedented events like the 2022 floods, provide a framework for detecting when something abnormal is happening - and for a country like Pakistan, where I have personal roots and where 30 million people were displaced in one summer, that kind of early warning could make a real difference.

# References

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